

ECE 647: Lecture 13

Continuity of Convex Functions

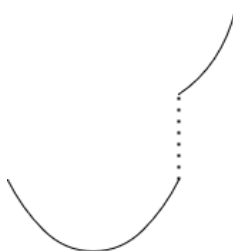
The simple property that:

$$f(\theta x + (1 - \theta)y) \geq \theta f(x) + (1 - \theta)f(y)$$

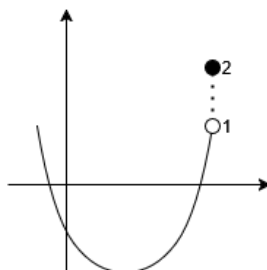
is in fact very strong and has important implications both geometrically and algebraically!

For example:

Let f be a convex function; then f is continuous on the interior of its domain.
Discontinuity cannot occur in the interior of the domain:



However, discontinuity can occur at the boundary of the domain:

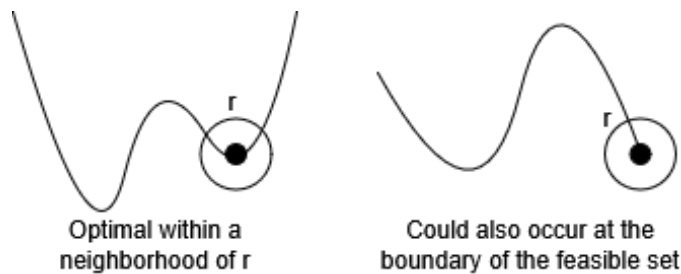


More on this when we discuss separation theorems.

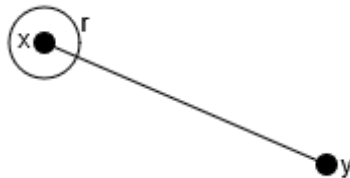
Basic Properties of Convex Problems

1. Every local optimum is also the global optimum.

- x is locally optimal if x is feasible and:
 $f_o(x) = \inf\{f_o(z) | z \text{ is feasible, } \|z - x\|_2 \leq r\}$ for some $r > 0$



- **Proof:** If $f(y) < f(x)$ for another feasible point y , then the line segment xy must lie in the feasible set, which is convex



By convexity of f , for any point

$$z = \theta x + (1 - \theta)y, 0 < \theta < 1$$

we must have

$$f(z) \leq \theta f(x) + (1 - \theta)f(y) < f(x)$$

Hence, there must exist a point in the neighborhood of x that has a smaller function value than $f(x)$.

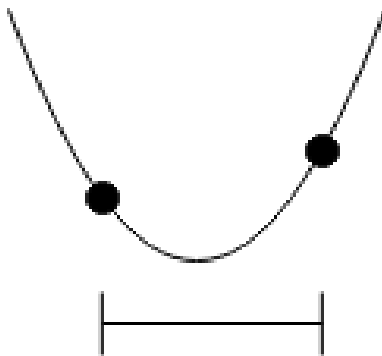
$\Rightarrow$ **This is a contradiction.**

- Note that the same conclusion holds for the looser definition of convex problems as well

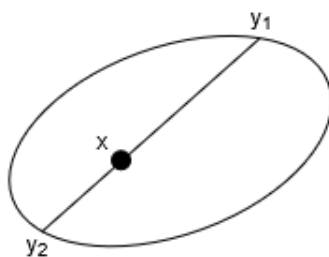
2. Necessary conditions for optimality are also sufficient. More details soon.

Maximizing Convex Functions

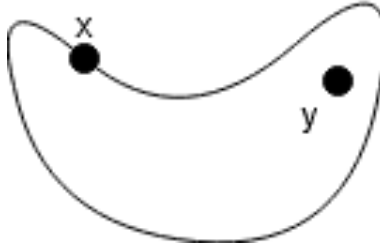
- Q: What if we maximize a convex function on a convex set?
- A: The maximum will be attained at the boundary.



- Both could be local optimums (but are not always)
- Let f be a convex function $C \rightarrow \mathbb{R}$. C is a convex set.
- Let D be a closed bounded set contained in C .
- $\Rightarrow$ for any $x \in \text{int}(D)$, there exists a $y \in \text{bd}(D)$ such that $f(y) \geq f(x)$.



- $f(x) \leq \max\{f(y_1), f(y_2)\}$
- $\Rightarrow$ Maximum on a closed set D must be attained at the boundary.
- Similarly, if you min a convex function on a non-convex set, then a local minimum at the boundary may not be the global minimum.



Standard Form

In this class, we will use the stricter form throughout, i.e. requiring f_i 's be convex, and h_i 's be linear.

Unfortunately, it can be difficult to tell for sure if a problem is a convex problem or not because even if the problem does not look like the standard form at first glance, it could be converted to the standard form. (Although some problems cannot be converted to the standard form.)

Clearly, we can always convert an optimization problem to a minimization problem. E.g. $\max\{f_o(x)\} \Rightarrow \min\{-f_o(x)\}$

Similarly, we can convert inequality constraints to $f_i(x) \leq 0$, and equality constraints to $h_i(x) = 0$.

However, the resulting f_o, f_i, h_i may not be convex or linear!

In these cases, additional tricks and experience will be very useful in identifying convex problems.

Transformation to a Convex Problem

- Change of variables

$$\begin{aligned} \min\{f_o(x)\} &\Rightarrow \min\{f_o(g(z))\} \\ \text{sub to } f_i(x) \leq 0 &\Rightarrow \text{sub to } f_i(g(z)) \\ \text{sub to } h_i(x) \leq 0 &\Rightarrow \text{sub to } h_i(g(z)) \\ &\text{need } g(\text{dom}(g)) \supseteq D \end{aligned}$$

- e.g.

$$\begin{aligned} \min\{x^2 + y\} \\ \text{sub to } \log(y + z) \leq x \\ y^2 + z^2 \leq x \end{aligned}$$

- **Example 1:** A constraint

$$\begin{aligned} & \log(y + z) \leq x \\ \Leftrightarrow & \log(y + z) - x \leq 0 \text{ **not convex** } \\ & \log(e^{y'} + e^{z'}) - x \leq 0 \text{ **convex** } \end{aligned}$$

- Caveat: Other functions of y and z may not be convex in y' and z' .

- **Example 2:**

$$\begin{aligned} & \min\{\sqrt{x_1} - \sqrt{x_2}\} \\ \text{sub to } & x_1 + x_2 \leq 1 \\ \Leftrightarrow & \min\{x'_1 - x'_2\} \\ \text{sub to } & x'^2_1 + x'^2_2 \leq 1, x'_1 \geq 0, x'_2 \geq 0 \end{aligned}$$

- **Transformation of functions**